Name _____

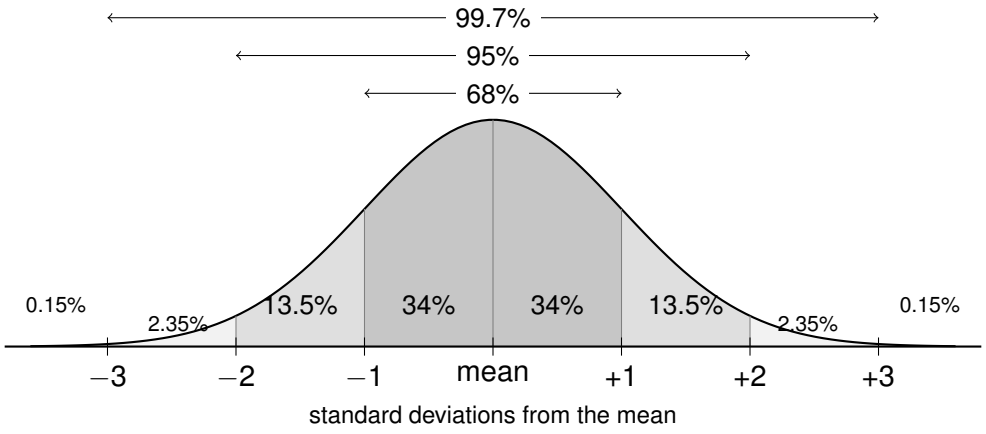
# Normal Distributions and the Empirical Rule KEY

Measure the heights of thousands of adult women. The histogram makes a smooth **bell shape**. Data like this has a **normal distribution**.

**A normal distribution:**

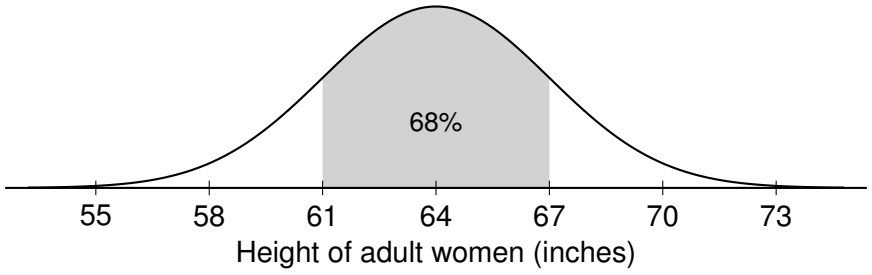
- is bell-shaped and symmetric
- has the mean and the median in the center (they are equal)
- is described by two numbers: the **mean** (center) and the **SD** (spread)

**The empirical rule (68-95-99.7 rule)**




About **68%** of the data is within **1 SD** of the mean.
About **95%** of the data is within **2 SDs** of the mean.
About **99.7%** of the data is within **3 SDs** of the mean.

**Example:** Heights of adult women have a mean of about 64 inches and an SD of about 3 inches.



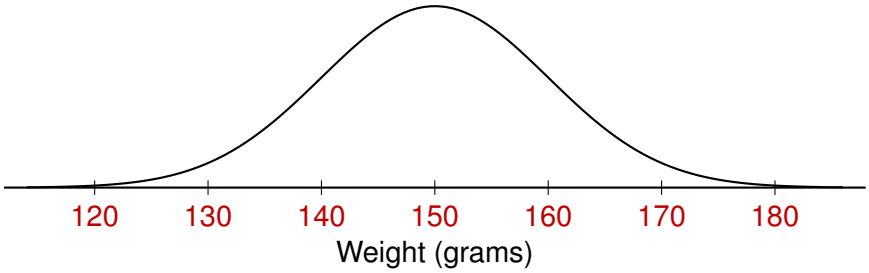

| | |
|---|---|
| Percent from 61 to 67 inches | 68% (within 1 SD) |
| Percent from 58 to 70 inches | 95% (within 2 SDs) |
| Percent taller than 67 inches | 13.5% + 2.35% + 0.15% = 16% |
| Percentile of 67 inches | 100% − 16% = 84%, the 84th percentile |

**Tip:** Draw the curve and label it every time. Then add up the parts you need.

Work with a partner. Compare answers when you finish each problem.

1. The weights of apples from one orchard have a normal distribution. The mean is 150 grams and the SD is 10 grams.

   Label the number line with the weights.




   About what percent of the apples weigh:

   (a) from 140 to 160 grams? 68%

   (b) from 130 to 170 grams? 95%

   (c) more than 160 grams? 13.5 + 2.35 + 0.15 = 16%

   (d) less than 130 grams? 2.35 + 0.15 = 2.5%

   (e) from 150 to 170 grams? 34 + 13.5 = 47.5%

   (f) from 130 to 160 grams? 13.5 + 34 + 34 = 81.5%

2. A phone battery lasts a mean of 20 hours with an SD of 2 hours. The times have a normal distribution.

   (a) Between what two times do the middle 95% of batteries last? $20 - 4 = 16$ to $20 + 4 = 24$ hours

   (b) What percentile is 22 hours? 50 + 34 = 84, the 84th percentile

   (c) About what percent of batteries last less than 16 hours? 2.5%

   (d) A battery lasts 27 hours. Is that unusual? Explain.
   Yes. 27 is 3.5 SDs above the mean. Almost no batteries (less than 0.15%) last that long.

3. In a normal distribution, what percentile is a value with each z-score?

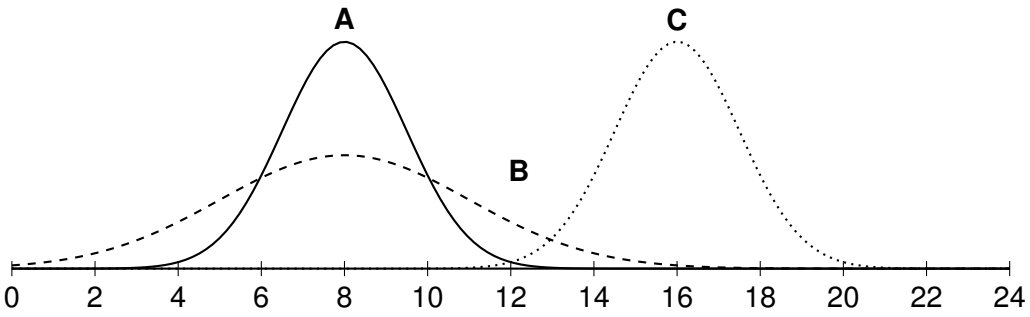
   (a) $z = 0$ 50th percentile

   (b) $z = 1$ 50 + 34 = 84th percentile

   (c) $z = -2$ 2.35 + 0.15 = 2.5th percentile

   (d) $z = 2$ 100 − 2.5 = 97.5th percentile

4. The three curves are normal distributions.




   (a) Which two curves have the same mean? A and B (both 8)

   (b) Which curve has the largest SD? Explain. B. It is the widest and most spread out.

   (c) How are A and C the same? How are they different?
   Same SD (same width and shape). Different means: A is centered at 8, C at 16.

5. Would each data set have a normal distribution? Explain.

   (a) The heights of all adult men in Missouri. Yes. Heights are bell-shaped and symmetric.

   (b) The yearly income of every household in the US. No. Income is skewed right.

   (c) Scores on a very easy test. No. Most scores are high, so it is skewed left.